

DATA SCIENTIST INTERVIEW TASK · JAN 2026

Identifying Students at Risk

A Proactive, Data-Driven Early
Intervention Model

🎯 The Objective

Proactively identify at-risk students before they disengage or withdraw

⚠️ The Challenge

Dirty data — inconsistencies, duplicates, and missing values across academic, attendance & financial records

⚡ The Goal

Shift from reactive damage control to proactive, targeted action plans

| Data Quality Issues & Solutions

Step 1 of 4 — Data Preparation

↔ Inconsistency

Problem

Mixed gender labels (e.g., 'Male' vs 'M') and inconsistent date formats across records

Solution

Standardised all labels to single-character codes and unified date formats (DD/MM/YYYY)

Fixed

⚠ Anomalies

Problem

Duplicate record for Student 1006; non-numeric value '..' in CreditsCompleted field

Solution

Removed duplicate rows; replaced invalid entries with NaN and flagged for risk scoring

Fixed

🔍 Missing Values

Problem

Null values in AttendanceRate and LastAttendanceDate — gaps that could hide at-risk behaviour

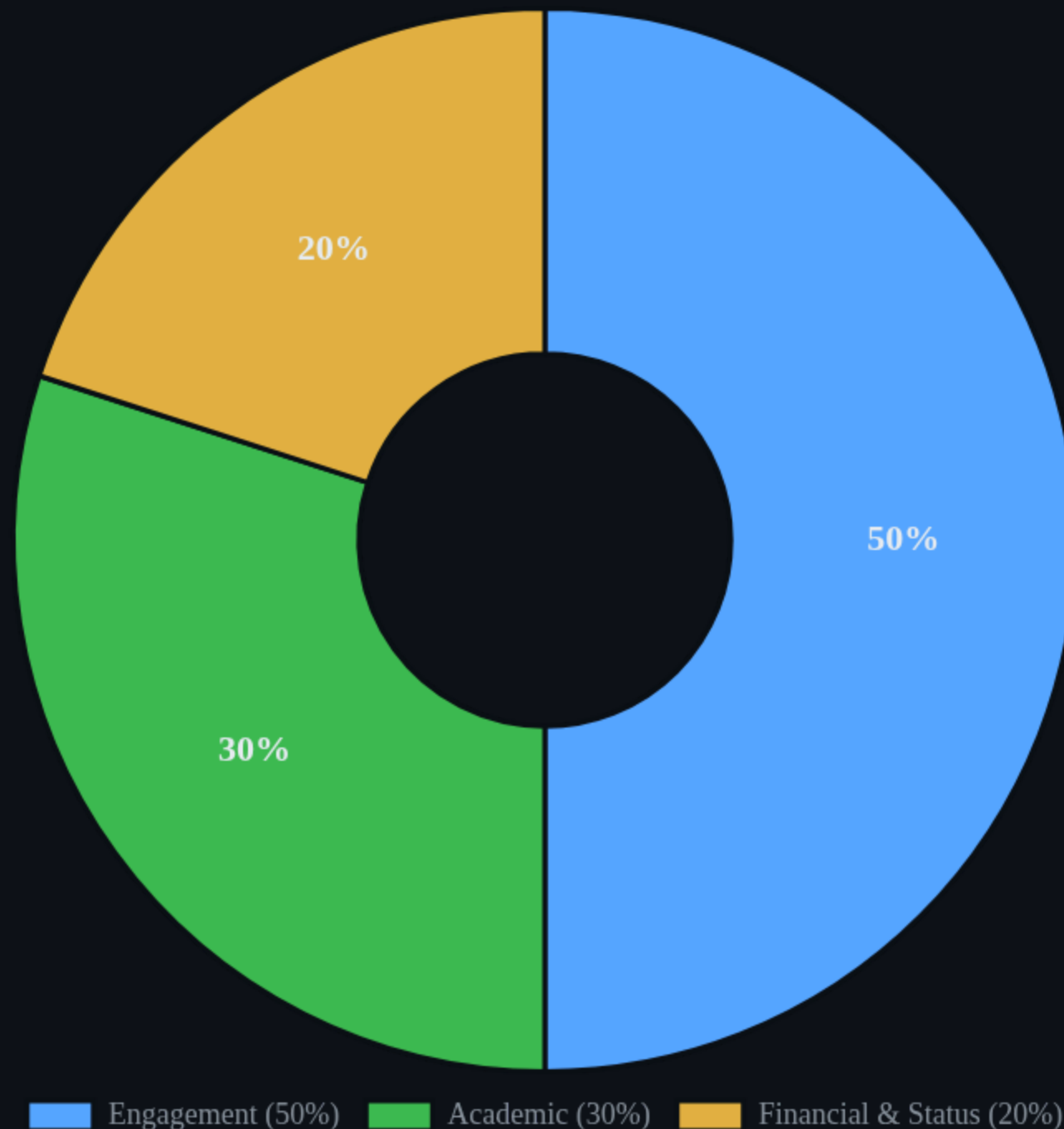
Solution

Missing data treated as a 'High Risk' signal — absence of data is itself a warning flag

Insight

Methodology — The Weighted Scoring Model Step 2 of 4 — Model Design

Score Composition



 **Engagement**
50% **AttendanceRate** **LastAttendanceDate**
Strongest lead indicator — disengagement precedes withdrawal

 **Academic**
30% **CreditsCompleted** **SubmittedLastAssessment**
Academic failure is a key non-completion driver

 **Financial & Status**
20% **OutstandingFees** **Scholarship** **EnrolmentStatus**
Financial stress and inactive status amplify drop-out risk

At-Risk Student Identification

Step 3 of 4 — Model Output

Student 1003 **Critical** 1.25 / 10

Status: Suspended Debt: £3,500 No attendance since 2025

Immediate financial & academic intervention required

Student 1002 **High** 2.75 / 10

Status: Active Attendance: 45% No recent logins

Outreach for engagement — high dropout probability despite active status

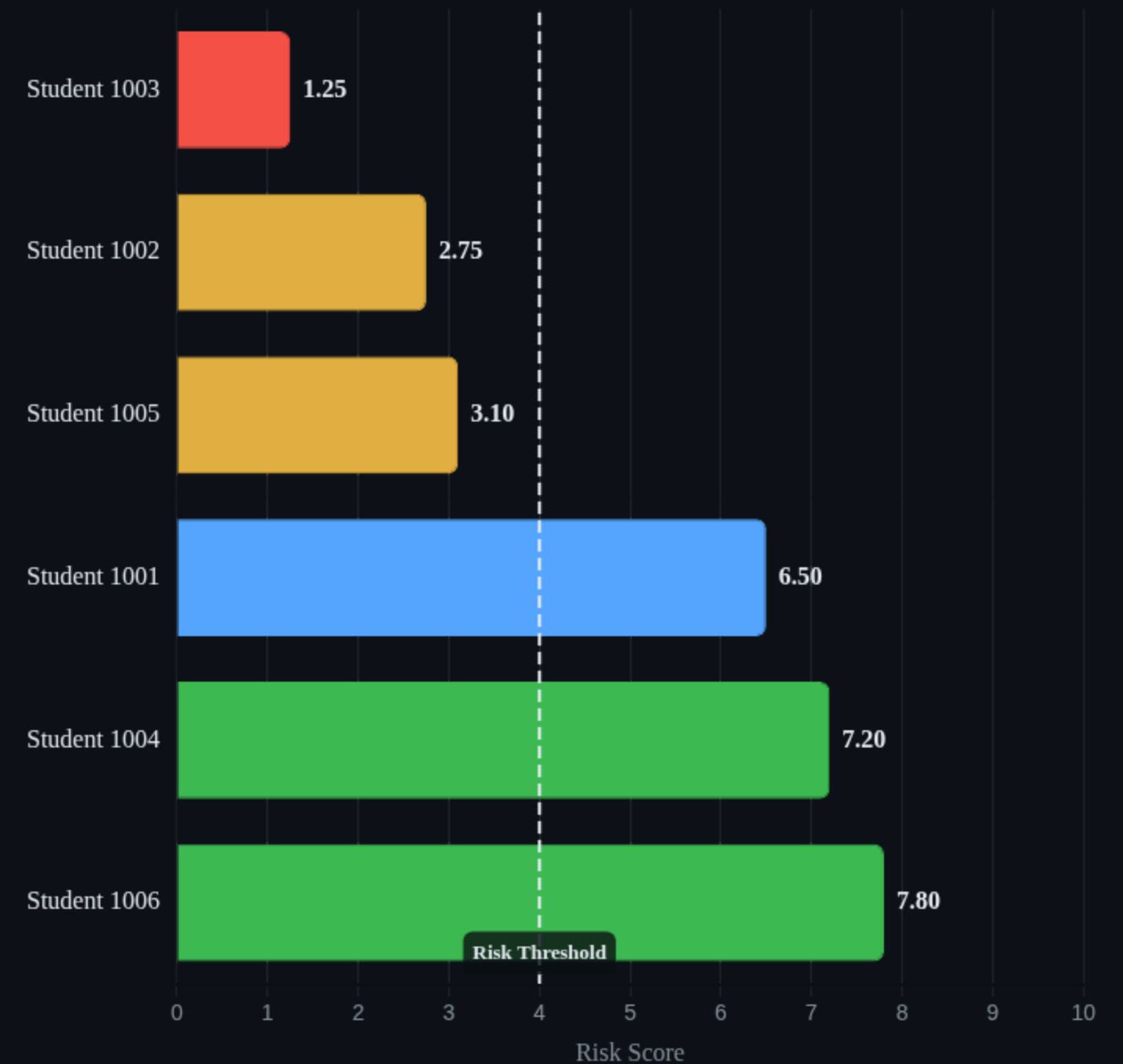
Student 1005 **Validation Case**

Known withdrawal Model correctly flagged Confirms model validity

Used as ground-truth benchmark to validate scoring logic

Risk Score by Student

Lower = Higher Risk



Limitations & Future Improvements

Step 4 of 4 — Reflection & Roadmap

⚠️ Current Limitations

 **Small Sample Size**
Snapshot data prevents detection of long-term trends or seasonal patterns

 **Missing Behavioural Data**
No VLE login frequency or engagement depth — key predictors of disengagement

 **Static Snapshot**
Risk scores are point-in-time; a student's situation may have already changed

🌱 Future Improvements

 **VLE Behavioural Data**
Integrate login frequency and resource access patterns as leading engagement indicators

 **Automated Live Dashboard**
Scores refresh automatically when a student misses a lecture or assessment deadline

 **Larger Longitudinal Dataset**
Train a predictive model on historical cohorts to improve score calibration and accuracy